

Heaven's Light is Our Guide



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Rajshahi University of Engineering & Technology, Bangladesh

Cancer Disease Prediction Using Data Mining Techniques

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CERTIFICATE

*This is to certify that, this thesis report entitled ” **Cancer Disease Prediction Using Data Mining Techniques,**” submitted by **Towhidul Islam**, Roll. **133086**, in partial fulfillment of the requirement for the award of degree Bachelor of Science in Computer Science & Engineering to Rajshahi University of Engineering & Technology, Rajshahi is a record of the candidate own work carried out by him under my supervision. This thesis has not been submitted for the award of any other degree.*

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ABSTRACT

Data mining techniques are a widely used tools in medical science for the diagnosis purpose of many diseases. This paper aims to develop an idea to diagnose cancer in early stages with better accuracy that helps to reduce the mortality rate of cancer death. Having the ability to generate understandable rules and handling capability of both continuous and categorical variables, decision tree is an effective approach that is applied on the datasets to diagnose cancer disease. But in terms of multi-class data with huge number of attributes, decision tree approach fails to predict the accurate classes more effectively. Later, an ensemble approach is presented to boost the decision tree via Adaptive Boosting technique that removes the limitations of decision tree. This approach is not applied previously on cancer related data field that generates result by combining multiple models where every model overcomes the error of the previous model and finally gives a result which is more accurate. Empirical result demonstrates the efficacy of the proposed Boosted Decision Tree classifier compared to baseline approach like Decision Tree.

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Chapter 1

INTRODUCTION

Introduction

Human body consists of many feedback loops which keep the biotic balance disability or malfunction of any of these loops causes severe diseases with short or long term complications. Data Mining techniques are considered as helpful methods for the diagnosis of many diseases. They, in fact, are probable models which have been proved useful in displaying complex systems and showing the relationships between variables in a graphic way. The advantage of this model is that it can take into account the uncertainty and can get the scenarios of the system change for the evaluation of diagnosis procedure. In this study Decision Tree and Boosting Algorithm are compared on several cancer data sets. Here, Decision Tree is used as a weak learner and Boosting Algorithm is used to make it as a strong learner. When Boosting Algorithm is applied on the result that is obtained by the Decision Tree then this procedure is called Boosted Decision Tree approach. The results indicated that the Boosted Decision Tree improves the performance which is more accurate than Decision Tree in Cancer Prediction.

1.1 Scope of Work

Cancer becomes the most dangerous cause of death for men and women now a days. Early detection and prediction can be a great idea to reduce cancer death. Increasing the awareness among the general people about cancer can also be a solution to reduce the percentage of death. There are too many complexities and uncertainties which compel us to use advanced controlling techniques.

1.2 Motivation

Cancer is the 6th leading cause of death and accounts for 10% of all mortality in Bangladesh. Based on two hospital-based cancer registries, nearly 66% of patients suffer from cancer within the age limit 30 to 65 [1].

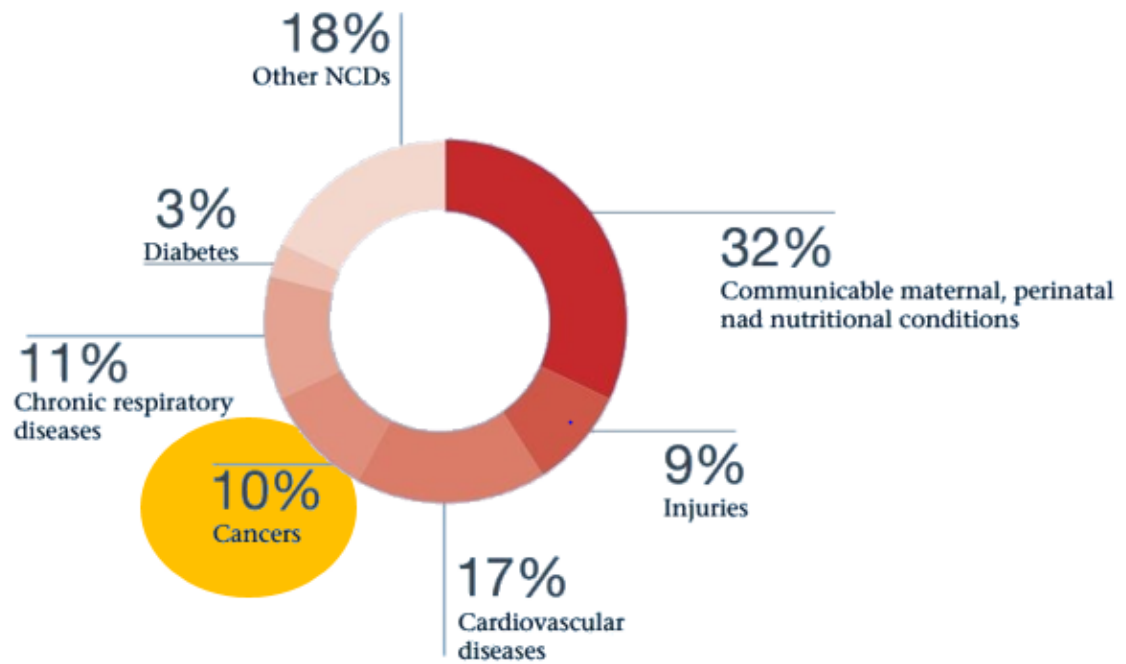


Figure 1.1: Proportional mortality (% of total deaths, all ages, both sexes) in Bangladesh [2]

Two most dangerous cancer that are the main reason for most of the mortality:

- Breast Cancer.
- Lung Cancer.

Men(%)	Women(%)	Both Sexes(%)
Lung(25.5%)	Lung(5.6%)	Lung(17.3%)
Lymphoma(7.4%)	Lymphoma(4.1%)	Lymphoma(6.0%)
Esophagus(5.9%)	Esophagus(3.4%)	Esophagus(4.6%)
Larynx(5.4%)	Breast(25.6%)	Breast(12.3%)
Stomach(5.1%)	Cervix uteri(21.5%)	Cervix uteri(9.1%)

Figure 1.2: Top five malignancies based on sex attending NICRH [2]

The annual mortality rate per 100,000 people from lung cancer in Bangladesh has increased by 4.0% since 1990, an average of 0.2% a year [3]. About 14% of all new cancers are lung cancer [4].

The World Health Organization(WHO) estimates approximately 1.38 million new breast cancer cases each year, resulting in 458,000 deaths annually. Among the total female population, 32.8 percent of female cancer patients suffering from breast cancer [5].

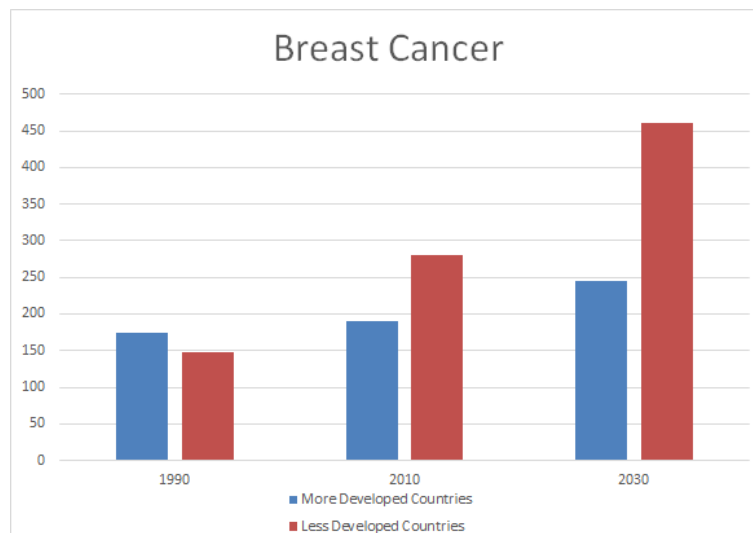


Figure 1.3: Breast Cancer Impact Relative to less and more developed countries [6]

From the figure, we can tell that in 1990s the mortality rate of breast cancer in more developed countries was higher than the less developed countries. But with the evaluation of time we can see that the mortality rate of breast cancer in more developed countries is lower than

the less developed countries.

As the time goes, more technologies are invented that are the main reason of reducing mortality rate of breast cancer in more developed countries. Data mining techniques are very helpful for predicting cancer because of there structures. Since cancer is a dangerous diseases which gives death to people. So, the motivation of this work was to reduce the mortal rate of cancer death and gave people a hope of living better life.

1.3 Objectives

- To improve Decision Tree using Boosting Algorithms.
- Implementation of Decision Tree and Boosted Decision Tree.
- Design an algorithm of Decision Tree and Boosted Decision Tree.
- Making a comparison between Decision Tree and Boosted Decision Tree.

1.4 Contributions

Our contributions for predicting cancer are as follows:

1. We have introduced Boosting algorithm for the purpose of predicting cancer diseases which is not applied before on cancer data set;
2. A new algorithm for predicting cancer diseases which can be applicable on both the multi-class and binary class of cancer dataset;
3. At first we have applied Decision Tree approach as a weak learner for the cancer prediction;
4. Finally, an proposed algorithm named as **Boosted Decision Tree** that improves the performance of Decision Tree and could help to diagnose cancer diseases having the fewest number of errors.

1.5 Organization of Thesis

Chapter 1 shows a short overview of cancer, motivation and organization of this research book.

Chapter 2 is dedicated for description about cancer prediction, details about data mining, detailed of applications, challenges and related works.

Chapter 3 is dedicated for the methodology of the system in details. Flowchart, Algorithm are described section by section. In the subsection, Testing and Training methods are described. Sample description about Ensemble Learning classifier is given.

Chapter 4 depicts the experimental evaluation of the system. Dataset, experimental setup are described in details in this chapter. Implementation procedure and cross-validation are also discussed in this chapter.

Chapter 5 is dedicated for the evaluation matrix, result and the performance analysis. It contains Comparison between the weak learner and our proposed algorithm in terms of standard metrics.

Chapter 6 represents the summary of this research work and highlights the overall contribution. A direction also given on the future scope of work by our proposed algorithm.

Problem Statement: Cancer Disease Prediction Using Data Mining Techniques.

To identify whether a given instance in dataset will be in the risk of cancer will be done on the basis of attribute values. Among the types of cancer, breast cancer and lung cancer are the main reason of the death for cancer.

So, we have used three datasets here to predict the risk of cancer diseases. Among them two are the breast cancer datasets and one is lung cancer dataset.

Breast cancer dataset contains all the details of instance like Clump Thickness, Uniformity of Cell Size, Uniformity of Cell Shape, Marginal Adhesion, Single Epithelial Cell Size, Bare Nuclei, Bland Chromatin, Normal Nucleoli, Mitoses, Age, BMI, Glucose, Insulin, HOMA, Leptin, Adiponectin, Resistin, MCP [7] [8].

Lung Cancer dataset contains all the details of the instance like Lingering, Coughing up phlegm or blood value, Hoarseness value, Shortness of breath value, Weakness and Fatigue value, Loss

of appetite and Weight value [9].

This attributes may contribute to identify whether a person is in the risk of cancer or not. Data Mining techniques like Decision Tree and our proposed algorithm Boosted Decision Tree are applied on these datasets to predict the type of cancer. The decision, which instance belongs to which cancer type will be taken based on the attribute values of the instances. A brief explanation has been given below through a flow chart.

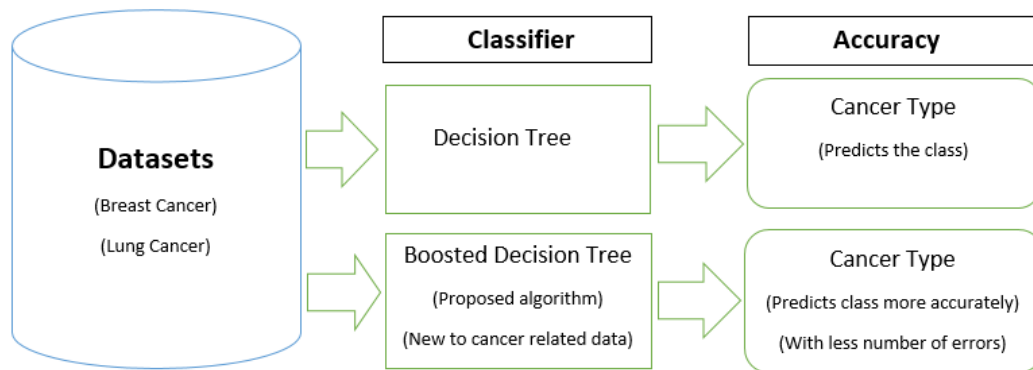


Figure 1.4: Flow chart of our proposed algorithm.

Data Mining:

Data mining(the analysis step of the "Knowledge Discovery in Databases" process, or KDD), an interdisciplinary subfield of computer science, is the computational process of discovering patters in large data sets involving methods at the intersection of artificial intelligence, machine learning, statistics and database systems. The over all goal of data mining process is to extract information from a data set and transform it into an understandable structure for further use. Aside from the raw analysis step, it involves database and data management aspects, data pre-processing, model and inference considerations, interestingness metrics, complexity considerations, post-processing of discovered structures, visualization and online updating [10].

It allows users to analyze data from many different dimensions or angels, categorize it and summarize the relationships identified. Technically, data mining is the process of finding cor-

relations or patterns among dozens of fields in large relational databases.

Decision Tree: [10]

- It is a supervised learning algorithm.
- Composed of set of questions or tests which will guide us to the destination.
- The motive of decision tree is to create a training model which can be used to predict class or values of target variables.

Ensemble Learning:

Ensemble means group. In ensemble learning , individual models come together and bring forth a model that is more accurate. In one word, multiple model coming together to give a final model [11]. There are two types of ensemble learning [12].

- Bagging.
- Boosting.

Boosting Algorithm:

Boosting algorithm is an ensemble learning algorithm. This algorithm is very effective in the diagnosis of many diseases because of their structures. The features of boosting algorithm are given below: [13]

- In case of boosting, the models are built in series.
- In each successive model, the weights are adjusted based on the learning of the previous model.
- Incrementally building an ensemble by training each new model instance.
- To emphasize the training instances that previous model miss-classified.

We have also planned to use this Boosting Algorithm as our proposed algorithm named **Boosted Decision Tree** because of it's structures of predicting many diseases effectively. Moreover, we have taken boosting algorithm as our proposed algorithm because it is a new on this cancer

related data field and also belonging the properties that can help to diagnose cancer diseases having the fewest number of errors.

Based on the properties of our proposed algorithm named **Boosted Decision Tree**, we have planned to use this on the Cancer datasets to predict the Cancer types in early stages. If we predict cancer in early stages, it will reduce the mortality rate of cancer death. This is the main issue of our research work.

In this study, the aim was to develop an idea which could help to diagnose cancer having the fewest number of errors. There was also an aim to help the doctors in early prediction of cancer diseases through this work. Decision tree and Boosting Algorithms are more effective in the diagnosis of many diseases because of their structures. Boosting algorithms are ensemble learning algorithms where multiple models are built and each model overcomes the error of the previous model. This way ensemble learning algorithms give better accuracy than a single model like Decision Tree.

Conclusion

It is important in order to make effective medical diagnosis. The aim of data mining is to extract knowledge from information stored in dataset and generate clear and understandable description of patterns. This study aims at the discovery of a decision tree model for the diagnosis of cancer. The datasets are collected from UCL Machine Learning Repository. Pre-processing is used to improve the quality of the data. The techniques, are pre-processing applied for attributes identification and section, data normalization and numerical discretization. Then Boosting Algorithm is used to improve the performance of Decision Tree. Boosting Algorithm gives result which is the combination of multiple models. In this algorithm, the error of the previous model is overcome in the next model and finally gives the combined result which could help to diagnose cancer diseases having the fewest number of errors.

Chapter 2

CANCER AND IT'S SYMPTOMS

Introduction

There are many diseases that lead people to death. Cancer is one of the most dangerous diseases that is responsible for the death of both men and women. So, early prediction can be a solution that reduces the mortality rate of cancer and gives people a hope of living better life. In this chapter we described definition of cancer, difference between cancer cell and normal cell, symptoms of cancer and types of cancer.

2.1 Definition of Cancer

Cancer is the uncontrolled growth of abnormal cells in the body. Cancer develops when the body's normal control mechanism stops working. Old cells do not die and instead grow out of control, forming new, abnormal cells. These extra cells may form a mass of tissue, called a tumor. Some cancers, such as leukemia, do not form tumors. Cancer may occur anywhere in the body. In women, breast cancer is one of the most common. In men, it's prostate cancer. Lung cancer and colorectal cancer affect both men and women in high numbers [14].

2.2 Difference Between Cancer and Normal Cell

Not every change in the body's tissues is cancer. Some tissue changes may develop into cancer if they are not treated, however. Here are some examples of tissue changes that are not cancer but, in some cases, are monitored [15]:

Hyperplasia occurs when cells within a tissue divide faster than normal and extra cells build up or proliferate. However, the cells and the way of the tissue is organized look normal under a microscope. Hyperplasia can be caused by several factors or conditions, including chronic irritation.

Dysplasia is a more serious condition than hyperplasia. In dysplasia, there is also a build up of extra cells. But the cells look abnormal and there are changes in how the tissue is organized. In general, the more abnormal the cells and tissue look, the greater the chance that cancer will form.

Some types of dysplasia may need to be monitored or treated. An example of dysplasia is an abnormal mole (called a dysplastic nevus) that forms on the skin. A dysplastic nevus can turn into melanoma, although most do not.

An even more serious condition is carcinoma in situ. Although it is sometimes called cancer, carcinoma in situ is not cancer because the abnormal cells do not spread beyond the original tissue. That is, they do not invade nearby tissue the way that cancer cells do. But, because some carcinomas in situ may become cancer, they are usually treated.

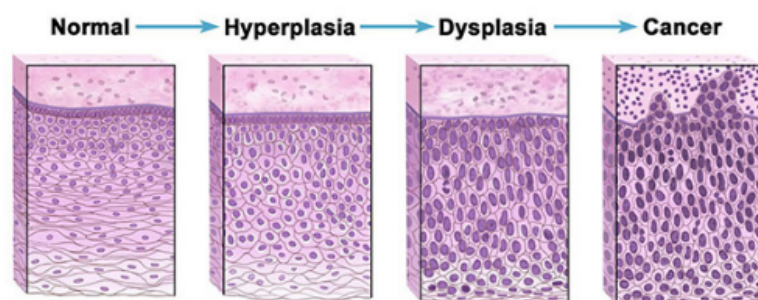


Figure 2.1: Difference between Cancer cell and Normal cell [15]

Normal cells may become cancer cells. Before cancer cells form in tissues of the body, the cells go through abnormal changes called hyperplasia and dysplasia. In hyperplasia, there is

an increase in the number of cells in an organ or tissue that appear normal under a microscope. In dysplasia, the cells look abnormal under a microscope but are not cancer. Hyperplasia and dysplasia may or may not become cancer [15].

2.3 Symptoms of Cancer

2.3.1 Symptoms for men

Cancer is among the most common causes of death in adult males in the U.S. While a healthy diet can lower the risk of developing certain cancers, other factors like genes can play a larger role. Once cancer spreads, it can be difficult to treat.

Knowing early symptoms can help to seek treatment sooner to better the chances of remission. Early symptoms of cancer in men include [16]:

- **Bowel changes:**

The occasional bowel problem is normal, but changes in bowels may indicate either colon or rectal cancer. These are collectively called colorectal cancers. Colon cancer can develop in any part of the colon, while rectal cancer affects your rectum, which connects the colon to the anus.

- **Rectal bleeding:**

Rectal bleeding may be an early sign of rectal cancer. This is especially concerning if the bleeding persists due to blood loss. Blood in stools can also be noticed for this type of cancer.

- **Urinary changes:**

Incontinence and other urinary changes may develop with the increasing of age. However, certain symptoms may indicate prostate cancer. Prostate cancer is most common cancer in men ages 60 and older. Common urinary symptoms include:

- Urinary leaks
- Incontinence
- An inability to urinate despite urges to go
- Delayed urination

– Straining during urination

- Blood in urine:

Having blood in urine is a common symptom of bladder cancer. This type of cancer is more common in current and former smokers than in people who've never smoked. Prostatitis, prostate cancer, and urinary tract infections can also cause blood in the urine. Early prostate cancer can also cause blood in the semen.

- Persistent back pain:

Back-pain is a common cause of disability, but few men realize that it may be a symptom of cancer. Symptoms of cancer may not show until it has spread to other parts of the body, such as the bones of the spine.

- Unusual coughing:

Coughing is not exclusive to smokers or to people with a cold or allergies. A persistent cough is an early sign of lung cancer. Coughing accompanied with bloody mucus is also associated with lung cancer in men.

- Testicular lumps:

Testicular cancers in men are less common than cancers of the prostate, lungs, and colon. Still, should not be ignored early symptoms. Lumps in the testicles are symptoms of testicular cancer. Doctors look for these lumps during wellness checks. For earliest detection, lumps should be checked once per month.

- Excessive fatigue:

Fatigue can be related to a number of chronic illnesses and medical disorders. Excessive fatigue tells that something goes wrong in your body. As cancer cells grow and reproduce, body may start to feel run down. Fatigue is a common symptom of various cancers. It is needed to go to doctor if excessive tiredness does not go away after a good night's sleep.

- Unexplained weight loss:

It becomes more difficult to maintain weight with the increasing of age, so weight loss can be considered as a positive thing. But sudden and unexplained weight loss can indicate a serious health problem, including almost any type of cancer.

- Lumps in breast:

Breast cancer is not exclusive to women. Men also need to be on guard and check for suspicious lumps in the breast area. This is the earliest detectable symptom of male breast cancer. Genes can play a role in male breast cancer, but it may also occur due to exposure to radiation or high estrogen levels. Breast lumps are most commonly found in men in their 60s.

2.3.2 Symptoms for women

Women's bodies are always changing. Sometimes changes that seem normal can be signs of cancer. A researcher named Robyn Andersen says about cancer to pay attention to the body so it might be noticed when something's different.

Knowing early symptoms can help to seek treatment sooner to better the chances of remission. Early symptoms of cancer in women include [16]:

- Breast changes:

Most breast lumps are not cancer but the doctors are needed to check them always. The women should know the changes too. The changes includes:

- Skin dimpling or puckering
- Nipples that turn inward
- Nipple discharge
- Redness or scaling of your nipple or breast skin

item Bloating:

An oncologist at NYU Langone Medical Center, Marleen Meyers says "Women are natural bloaters." It's OK to wait a week or two to see if it goes away. If the symptoms don't get better with time or happening weight loss or bleeding, it is needed to go under doctor's instruction. Constant bloating could be a sign of cancer, including breast, colon, gastrointestinal, ovarian, pancreatic, or uterine.

- Between-Period Bleeding:

Bleeding that's not a part of your usual monthly cycle, it can have many causes. Bleeding after menopause is never normal and should be checked right away.

- **Skin Changes:**

A change in the size, shape, or color of a mole or other spot, as well as development of new spots, are common signs of skin cancer.

- **Blood in Your Pee or Stool:**

Bloody stool is often from hemorrhoids but it can also be a symptom of colon cancer. Bloody urine is usually the first sign of cancer of the bladder or kidneys.

- **Changes in Lymph Nodes:**

Lymph nodes are small, bean-shaped glands around the body. Most changes in them come from common infections. But some cancers, including leukemia and lymphoma, can also cause lymph nodes to swell and/or become tender.

- **Trouble Swallowing:**

Occasional trouble swallowing is nothing to worry about. But when it happens often, especially with vomiting or weight loss, it can be a symptoms of throat or stomach cancer.

- **Weight Loss Without Trying:**

Most women wish extra pounds would magically melt away. But losing 10 pounds or more without a change in your diet or exercise habits could signal a problem. Most unintended weight loss is not cancer. A researcher, Meyers says. "It's often caused by stress or your thyroid, but it can be a sign of pancreatic cancer". Other types of cancer such as colon, stomach and lung cancers are also possible for losing weight.

- **Heartburn:**

Too much food, alcohol, or stress (or all three) can cause serious heartburn. Heartburn that doesn't go away or gets worse could mean cancer of the stomach, throat or ovaries. Also, persistent heartburn can damage the lining of esophagus and lead to a condition called Barrett's esophagus.

- **Fever:**

A fever that doesn't go away and can't be explained could mean leukemia or another blood cancer.

- **Belly Pain and Depression:**

It's rare, but belly pain plus depression can be a sign of pancreatic cancer.

2.4 Types of Cancer

Cancer is the most dangerous cause of death for men and women. The early detection and prediction of cancer can be helpful in curing the diseases completely. The following table gives the estimated numbers of new cases and deaths for each common cancer type:

Cancer Type	Estimated New Cases	Estimated Deaths
Bladder	81,190	17,240
Breast (Female – Male)	266,120 – 2,550	40,920 – 480
Colon and Rectal (Combined)	140,250	50,630
Endometrial	63,230	11,350
Kidney (Renal Cell and Renal Pelvis) Cancer	65,340	14,970
Leukemia (All Types)	60,300	24,370

Figure 2.2: Estimated numbers of new cases and deaths for each common cancer type [17]

From the figure we can see that Breast Cancer is most important reason for deaths of the women among all types of cancer. Again, according to the American Cancer Society 2018, Lung Cancer is the second most important reason for deaths in which estimated numbers of cases are 234,030 and the estimated numbers of deaths are 154,050 [17].

With the development of clinical technologies, different tumor features have been collected for Breast Cancer diagnosis. The objective of this research is to diagnose Breast Cancer based on the extracted tumor features. To extract useful information and diagnose the tumor, a hybrid of K-means and support vector machine (K-SVM) algorithms is developed. The K-means algorithms is utilized to recognize the hidden patterns of the Benign and Malignant tumors separately. The membership of each tumor to these patterns is calculated and treated as a new feature in the training model. Then, a support vector machine (SVM) is used to obtain the new classifier to differentiate the incoming tumors. Traditionally, Breast Cancer was predicted based on the mammography by radiologists and physicians [18].

Lung Cancer is one of the leading cause of cancer deaths in both women and men. Manifesta-

tion of Lung Cancer in the body of the patients reveals through early symptoms in most of the cases [19]. Symptoms that may suggest Lung Cancer includes: [20]

- Dyspnea(Shortness of breath with activity),
- Hemoptysis (coughing up blood),
- Chronic coughing or change in regular coughing pattern,
- Wheezing,
- Chest pain or pain in the abdomen,
- Cachexia (weight loss, fatigue, and loss of appetite),
- Dysphonia (hoarse voice),
- Clubbing of the fingernails(uncommon),
- Dysphasia(difficulty swallowing),
- Pain in shoulder, chest, arm,
- Bronchitis or pneumonia,
- Decline in Health and unexplained weight loss.

Mortality and morbidity due to tobacco use is very high. Usually lung cancer develops within the wall or epithelium of the bronchial tree. But it can start anywhere in the lungs and affect any part of the respiratory system.

There are two major types of lung cancer. They are Non-small cell lung cancer (NSCLC) and small cell lung cancer (SCLC) or oat cell cancer. Each type of lung cancer grows and spreads in different ways and is treated differently. If the cancer has features of both types, it is called mixed small cell/large cell cancer [20].

Non-small cell lung cancer is more common than SCLC and it generally grows and spreads more slowly. SCLC is almost related with smoking and grows more quickly and form large tumors that can spread widely through the body. They often start in the bronchi near the center of the chest. Lung cancer death rate is related to total amount of cigarette smoked [21].

Conclusion

Cancer is a fatal diseases. Among the types of cancer, breast cancer and the lung cancer are more responsible for the mortality of men and women. There are many symptoms of cancer whom that the types of cancer can be predict primarily. During the past two decades, many such prediction models have been developed. This models are applied applied on the datasets and predict which attributes are responsible for the cancer type. Cancer leads death to the people at early ages. So, early prediction can save many lives. In this chapter, we have discussed the cancer definition, cancer types, their symptoms moreover the background study related our thesis work.

Chapter 3

LITERATURE REVIEW

Introduction

For the working purpose, many related articles, papers, documents have been read to achieve the knowledge. This documents are very helpful to enrich our ideas, efficiency and potentiality that has been used to do our thesis. The articles that help us to generate ideas and do our thesis work are given in the following sections.

3.1 Data

Data is a collection of qualitative or quantitative facts. In computing, data is information that has been translated into a form that is efficient for movement or processing. Relative to today's computers and transmission media, data is information converted into binary digital form. It is acceptable for data to be used as a singular subject or a plural subject [10].

3.2 Classification

Classification is a data mining function. It assigns items in a collection to target classes. The goal of classification is to accurately predict the target class for each case in the data [22].

3.3 Data Mining

Data Mining can be defined as [23]:

- Extraction of interesting(non-trivial, implicit, previously unknown and potentially useful) patterns or knowledge from huge amount of data.
- The discovery of hidden structures in large datasets.
- Methods to extract data patterns.
- Mostly used by the companies to turn raw data into useful information.

Goal of Data mining [23]:

- Extraction of information from a data set.
- Transformation of data into an understandable structure for further use.

Data mining(the analysis step of the "Knowledge Discovery in Databases" process, or KDD), an interdisciplinary subfield of computer science, is the computational process of discovering patterns in large data sets involving methods at the intersection of artificial intelligence, machine learning, statistics and database systems [10].

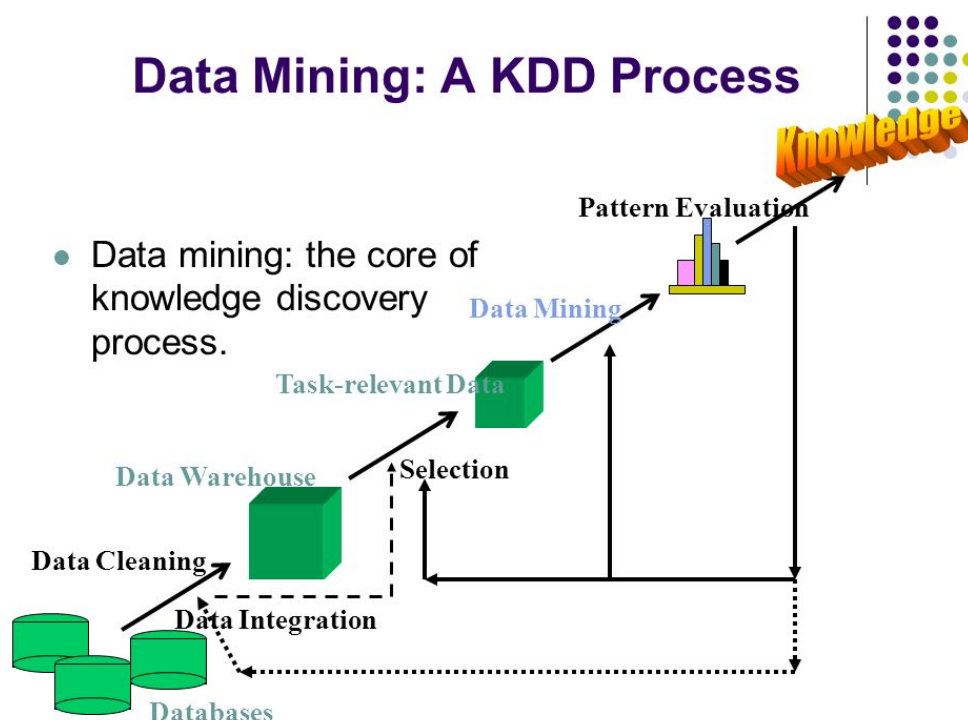


Figure 3.1: KDD process of data mining [10]

It allows users to analyze data from many different dimensions or angles, categorize it and summarize the relationships identified. Technically, data mining is the process of finding correlations or patterns among dozens of fields in large relational databases.

3.4 Medical Data Mining

Medical data mining plays an important role for the prediction of diseases in early stages. If the diseases are predicted early, it can be very helpful to reduce the mortality rate.

3.4.1 Medical Data

Medical data is considered as health related information that is associated with regular patient. Data mining holds great potential for the health care industry to enable health systems to systematically use data and analytics to identify inefficiencies and best practices that improve care and reduce costs. Some experts believe the opportunities to improve care and reduce costs concurrently could apply to as much as 30% of overall health care spending. This could be a win/win overall. But due to the complexity of health care and a slower rate of technology adoption, our industry lags behind these others in implementing effective data mining and analytic strategies [24].

3.4.2 Forms of Medical Data

Medical data can be in the following forms [24]:

- Administrative data.
- Claims data.
- Patient/disease data.
- Health surveys data.
- Clinical trial data.

3.4.3 Data Mining Approaches on Medical Data

Medical data mining can be defined as the following [25]:

- Accumulation of large quantities of information about patients.
- Technological advancements in the form of computer-based patient record.
- Evaluation of stored clinical data. .

The working strategies of medical data mining are indicated as follows:

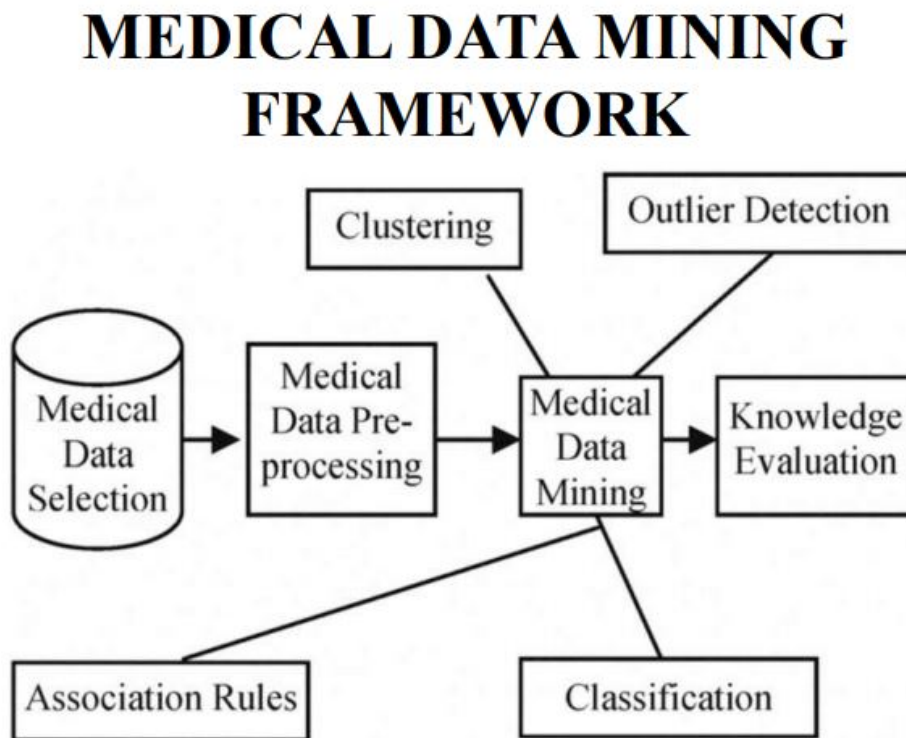


Figure 3.2: Medical Data Mining Process [25]

3.4.4 Applications of Medical Data Mining

- Increasing the efficiency and elimination of the human factor.
- Reduction of time and cost.
- Medical decision support system.
- Knowledge extraction.

3.5 Related Works

A research paper titled "Using Machine Learning Algorithms for Breast Cancer Risk Prediction and Diagnosis", showed the performance comparison among different Machine Learning Algorithms in Breast Cancer Prediction. Here Naive Bayes, support vector machine(SVM), Decision Tree, K Nearest Neighbors methods are used to obtain the accuracy. The limitation of this paper is that there is no information about the effect of linear and non-linear SVM [26].

Another research paper titled "Predicting Breast Cancer Survivability: a Comparison of three Data Mining Methods", proposed a Breast cancer Survivability Prediction model. Here Decision Tree, Simple Logistic Classification, Artificial Neural Network methods are used to predict the survivability model. The limitations of this paper is that other machine learning algorithms like SVM and Naive Bayes has not been used [27].

Another research paper titled "Data Mining Techniques: To Predict and Resolve Breast Cancer Survivability" made prediction on patient survivability. Rep Tree, RBF Network Model, Simple Logistic Regression methods are used for this purpose. Limitation of this paper is that there is no information about the degree of polynomial features for logistic classification [28].

Another research paper titled "Diagnosis of Lung Cancer Prediction System Using Data Mining Classification Techniques", proposed an effective model for early detection and correct diagnosis of lung cancer. Here Decision Tree, Naive Bayes, Artificial Neural Network methods are used to predict the model. Limitation of this paper is that there is no use of continuous data, only the categorical data are used for the prediction purpose [20].

Conclusion

In this chapter, we have discussed the literatures, articles and documents that need to implement our thesis work.

Chapter 4

PROPOSED METHODOLOGY AND IMPLEMENTATION

Introduction

In this chapter we will take a look at the implementation procedure and the proposed methodology, cross validation, steps of applying our proposed model on the data sets. Proposed methodology includes description about the base learner and our proposed algorithm to improve the performance of the base learner and make it as strong learner.

Implementation procedure includes experimental setup, dataset description and the partitioning procedures of data to find the desired output.

4.1 Proposed Methodology

4.1.1 Base Classifier: Decision Tree

Decision Tree is a supervised learning algorithm composed of set of questions or tests which guides to the destination. The motive of decision tree is to create a training model which can be used to predict class or values of target variables [10]. Because of this properties Decision Tree is used widely for the purpose of medical diagnosis. Decision Tree can be grown from the following steps [29]:

- Order the feature based on information gains(IG).
- At each step pick the feature with highest information gain(IG).

i.e. $A^* = \text{Arg max IG}(y, A_i)$

Where, Y = the given output label.

A_i = the given feature.

- Split the dataset based on A^* .
- Continue splitting until the partial datasets have homogeneous labels.

In order to apply Decision Tree approach on the cancer datasets the following equations have to be considered:

- Entropy calculation: Entropy is a measure regarding how “innovative” something is. The amount of uncertainty regarding something is. If X is a random variable, then entropy of X denoted by $H(X)$ is:

$$H(X) = - \sum_{i=a}^A P(A) \cdot \log_2 a \quad (4.1)$$

Where, A = alphabet and $a = \text{symbol} \in A$

- Specific Conditional Entropy Calculation: Entropy of Y given X , i.e, $H(Y/X)$ means finding the uncertainty in Y after knowing X

$$H(Y/X = a) = \sum_{i=a}^A P(a) \cdot H(Y/X = a) \quad (4.2)$$

Where X and Y are two random variables.

A is an alphabet of random variable X .

$X \rightarrow A \rightarrow \{a, b\}$

- Information Gain Calculation(I.G):

$$I.G(Y; X) = H(Y) - H(Y/X) \quad (4.3)$$

when $H(Y) = H(Y/X)$, then $I.G = 0$, means that X and Y are two statistically independent variables.

When $H(Y) > H(Y/X)$, then $I.G = \text{positive}$.

When $H(Y) < H(Y/X)$, then $I.G = \text{negative}$.

Decision Tree is one of the data mining techniques that is very helpful for predicting disease like cancer because of its structure. Decision Tree approaches are the most widely used data mining methods for several reasons in medical diagnosis [30]. It has the ability to generate understandable rules and to handle both continuous and categorical variables which is very effective to diagnose cancer diseases [31]. But in terms of multi-class data with huge number of attributes Decision Tree approach fails to predict the accurate classes more effectively [32]. To eliminate this problem, a number of data mining techniques have been suggested including SVM, Naive Bayes classifier that perform well where all of the techniques creates single model [33].

4.1.2 Boosting Algorithm

Boosting algorithm is the point of interest of this work because it is a new on this cancer related data field and also belonging the properties that can help to diagnose cancer diseases having the fewest number of errors [34].

Boosting Algorithm is a strong ensemble-based learning algorithm that iteratively improves the classification accuracy using any base learner [35]. Since boosting algorithm advances the classification performance, it has been used to enhance Decision Tree classifier performance on cancer related data sets [36] [37].

To generate models, Decision Tree has been used as a weak learner instead of a classifier to be boosted with AdaBoost [38] algorithm and after combining these models, a final model is generated. The method has been tested for three cancer data sets. Among them two are breast cancer data sets and one is lung cancer data set.

4.1.3 Adaboost

Adaptive Boosting is an ensemble learning algorithm. Where adaboost M1 is used for binary class prediction and adaboost M2 is used for multiple class prediction. The working procedure of adaptive boosting algorithm is as follows:

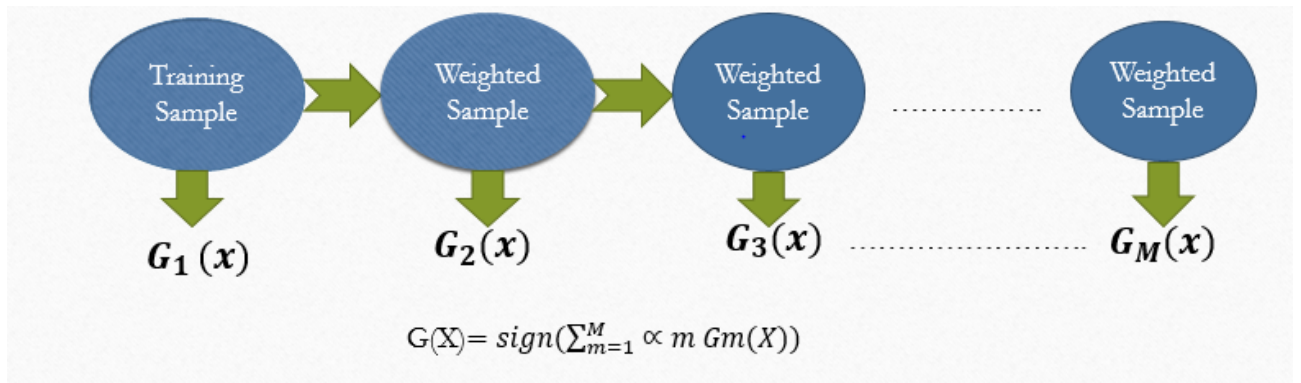


Figure 4.1: Working Process of Adaptive Boosting.

Here is an example how adaptive boosting algorithm works step by step [39]:

- Step 1:

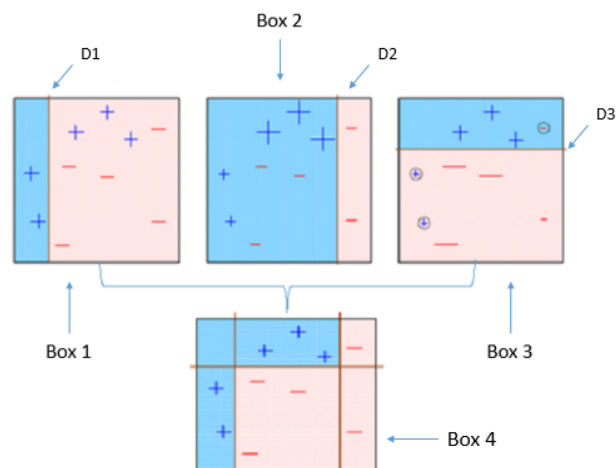


Figure 4.2: View of Dataset after 1st Step.

- Step 2 and 3:

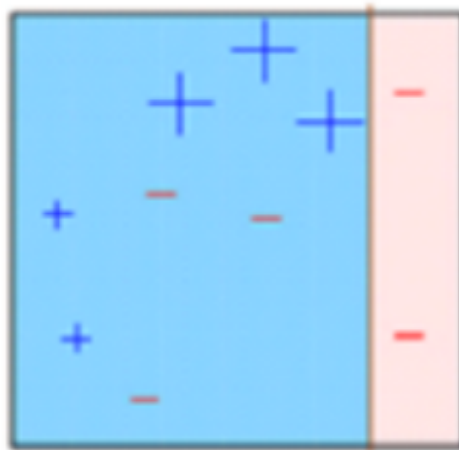
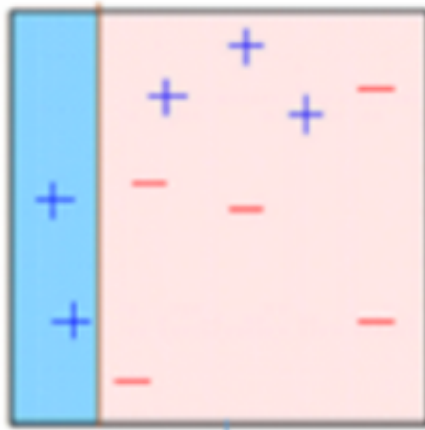


Figure 4.3: View of Dataset after the 2nd and 3rd Step.

- Step 4:

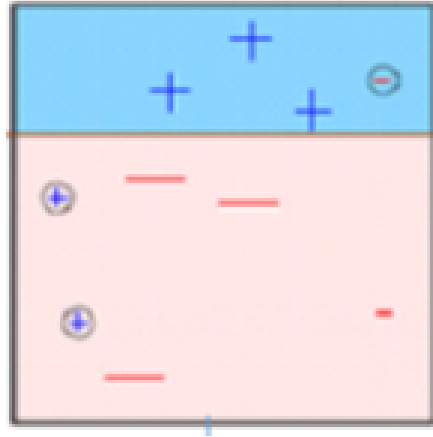


Figure 4.4: View of Dataset after 4th Step.

- Step 5:



Figure 4.5: Final View of the dataset after combination of all the Steps.

The AdaBoost algorithm considers a weak learner DT as a black box. A n points datasets $(x_1, y_2) \dots (x_n, y_n)$ are the inputs for the AdaBoost algorithm that provide the labels $y_i \in \{-1, 1\}$ for all i . It executes for M times and invokes the DT on each of its turns. New sets of input points are fed to DT to achieve new information from DT on each round. New classifications D_m are

initialized over the n input points at each round m as n is fixed and a sample of polynomial size is selected from new classification to give as input to DT. At each round, a function or model is generated by DT that outperforms than random guessing with respect to the particular classification D_m . After combining all the specialist weak models of each turn, a final model is generated which are built on hard samples [40].

4.1.4 Proposed Boosted Decision Tree Approach

This paper considers Decision Tree as the weak learner model that has been tried to make strong model through the proposed approach. Initially all of the input patterns are given equal priorities by the proposed approach. After the first model generation, the misclassified patterns are weighted higher than those that are truly classified so that the misclassified patterns can get higher probability of selection for the remaining model generations. In this proposed approach, the error of the previous model is overcome in the next model and finally gives the combined result which could help to diagnose cancer diseases having the fewest number of errors. This procedure continues until the calculated error is either 0 or ≥ 0.5 . The Boosted Decision Tree algorithm is summarized in **Algorithm 1**.

Algorithm 1. Boosted Decision Tree

1. Input: n training examples $\{x_i, y_i\}$, Where $x_i \in X$, $y_i \in \{1, \dots, y\}$, $i = 1, \dots, n$; # iterations M
2. Initialize the weights $w_i^1(x_i, y_i) = \frac{1}{n}$, Where, $i = 1, \dots, n$
3. For each round $m = 1, \dots, M$
 - Draw a sufficiently large sample from the classification D_m
 - Run Decision Tree approach on this sample to produce weak learner DT using the steps as follows:

Compute Entropy by the following Operations:

* If X is a random variable, then entropy of X denoted by $H(X)$ is:

$$H(X_i) = - \sum_{i=a}^A P(A_i) \cdot \log_2 a$$

Where, A = alphabet, $a = \text{symbol} \in A$.

Compute Specific Conditional Entropy by the following operations:

* Entropy of Y given X means, $H(Y/X)$ can be calculated by,

$$H(Y_i/X_i = a) = \sum_{i=a}^A P(a).H(Y_i/X_i = a)$$

Compute Information Gain(I.G) by the following operations:

* Information Gain

$$I.G(Y_i; X_i) = H(Y_i) - H(Y_i/X_i)$$

• Set err_m

$$err_m = \frac{\sum_1^N w_i^{(m)} I(y_i \neq G_m(x_i))}{\sum_1^N w_i^{(m)}}$$

• Set Θ_m

$$\Theta_m = \log \frac{(1 - err_m)}{err_m}$$

• Update Weights,

$$w_i^{(m+1)} = w_i^{(m)} . e^{[\Theta(m) I(y_i \neq G_m(x_i))]}$$

4. Output of Combined Model,

$$G(x) = \text{sign}(\sum_1^M \Theta_m . G_m(x))$$

4.2 Cross Validation

In statistical prediction problem, cross-validation methods act as standard to estimate the performance and effectiveness of classifiers. Most popular three cross-validation are: independent dataset test, jackknife test and k-fold cross-validation test. The K- fold cross-validation is a method to approximately estimate prediction error without bias under much more complicated situations.

In k-fold cross-validation, the original sample is randomly partitioned into k equal size subsamples. Of the k subsamples, a single subsample is retained as the validation data for testing the model, and the remaining k-1 subsamples are used as training data. The cross-validation process is then repeated k times (the folds), with each of the k subsamples used exactly once as the validation data. The k results from the folds can then be averaged (or

otherwise combined) to produce a single estimation. The advantage of this method is that all observations are used for both training and validation, and each observation is used for validation exactly once. Figure 4.6 shows a sample of 4-fold cross validation.

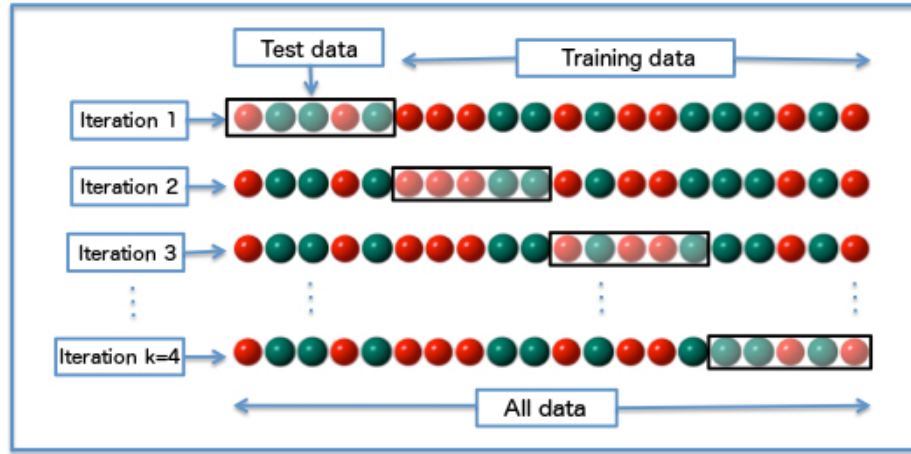


Figure 4.6: Diagram of k-fold cross-validation with $k = 4$ [41]

In our study, we have used 5-fold cross validation on our datasets. While 4 of the folds are for training and another is for testing.

4.3 Steps of Applying Proposed Model

- Load the data set to an array.
- Split the dataset into two partition. One is to train and another is to test.
- Apply decision tree and our proposed algorithm, boosted decision tree on the dataset.
- Measure Accuracy in terms of standard metrics.
- Compare the two and conclude a decision which one performs better.

4.4 Implementaion

4.4.1 Experimental Setup and Partitioning Procedure of Datasets

Fivefold cross-validation has been used to achieve the performance of the two mentioned methods, where each dataset is arbitrarily separated into five equal sized subdivisions. Each of the

five subdivisions was used precisely once as the testing dataset while the rest were in corresponding training dataset. Then the results of the five folds are averaged to appraise the final performance. MATLAB 16 has been used for the coding purpose of the algorithms in order to measure the performance. The machine, HP Pavilion 14-e036tx notebook pc[] has been used for the experimental purpose occupies; processor: Intel(R) Core(TM) i7-4702MQ @ 2.2GHz; RAM: 4.00 GB 1600 MHz DDR3, system type: 64-bit operating system.

4.4.2 Dataset Description and approach to find result

In this experiment four data sets are used which all are from UCL Machine Learning Repository [7] [8] [9] [42]. Three of them are the breast cancer datasets and one is lung cancer dataset. Our proposed algorithm is applied on this datasets. A combined result of multiple models is obtained which is more effective than the decision tree to diagnose cancer with the fewest number of errors. In every model this algorithm overcomes the error of the previous model and finally gives a result which is more accurate. Table 4.1 summarizes the properties of the selected data sets.

Table 4.1: Cancer Datasets.

Datasets names	No. of Instances	No. of Attributes	No. of Class Types
Breast Cancer Wisconsin	699	10	2
Breast Cancer Coimbra	116	10	2
Lung Cancer	32	57	3
Breast Cancer (Categorical)	286	9	2

Conclusion

In this chapter we have discussed the partitioning procedures of data sets, experimental set up, datasets description and the procedure how to apply our proposed algorithm on the datasets.

Chapter 5

RESULT ANALYSIS AND DISCUSSION

Introduction

Result analysis and performance evaluation are the main focus of a research work. It's also the main target of our work to increase the accuracy and show a good performance. Here we have used Ensemble Learning techniques which is not previously applied on cancer related data. In this chapter, we described about the evaluation matrices those are used for performance evaluation of our proposed algorithm **Boosted Decision Tree**.

5.1 Evaluation Metrics

In our work, well-defined metrics are used to measure the performance of our algorithm.

5.1.1 Accuracy (Acc)

Accuracy (Acc) is the ratio of a total number of correctly classified samples (C) and the total number of samples (N). It varies between 0(%) (least accurate) and 100(%) (most accurate). If the accuracy is 100 that means the predictor is best.

For calculating the performance of our proposed algorithm named **Boosted Decision Tree**, we have applied 5-fold cross-validation. For calculating the accuracy of our proposed algorithm, we have used Equation 5.1.

$$Acc = \frac{C}{N} \quad (5.1)$$

5.1.2 Precision (Pr)

Precision is defined by the fraction of retrieved documents that are relevant to the query. It also varies between 0 (least accurate) and 1 (most accurate). If the accuracy is 1 that means the predictor is best [43].

For calculating the precision of our proposed algorithm, we have used Equation 5.2.

$$Pr = \frac{|\{Relevant_items\} \cap \{Retrived_items\}|}{|\{Retrived_items\}|} = \frac{TP}{TP + FP} \quad (5.2)$$

5.1.3 Recall (Re)

Recall is defined by the fraction of the relevant documents that are successfully retrieved. It also varies between 0 (least accurate) and 1 (most accurate). If the accuracy is 1 that means the predictor is best [43].

For calculating the recall of our proposed algorithm, we have used Equation 5.3.

$$Re = \frac{|\{Relevant_items\} \cap \{Retrived_items\}|}{|\{Relevant_items\}|} = \frac{TP}{TP + FN} \quad (5.3)$$

5.1.4 F1 Score

F1 Score is measure of a test's accuracy. It considers both the precision, **Pr** and the recall, **Re** of the test to compute the **F1 Score**. **Pr** is the number of correct positive results divided by the number of all positive results returned by the classifier, and **Re** is the number of correct positive results divided by the number of all relevant samples [44].

For calculating the f1 score of our proposed algorithm, we have used Equation 5.2, 5.3 and 5.4.

$$F_1Score = 2 \frac{Precision * Recall}{Precision + Recall} \quad (5.4)$$

5.2 Experimental Result Analysis

There are four datasets of cancer on which we have applied our proposed algorithm **Boosted Decision Tree**. We have also taken **Decision Tree** as a weak learner and applied on this datasets and made a comparison between this two.

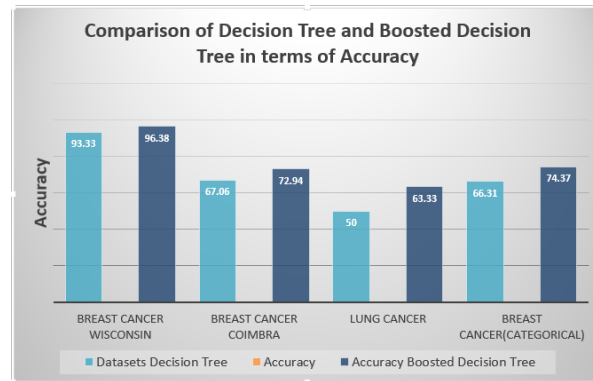


Figure 5.1: Graphical Comparison of Decision Tree and Boosted Decision Tree in terms of Accuracy.

Figure 5.1 shows the graphical comparison of Decision Tree and Boosted Decision Tree in terms of Accuracy for all of the four datasets. Here we can see that our proposed algorithm **Boosted Decision Tree** performs far better than the base learner **Decision Tree**.

Since accuracy is not a standard metric for performance analysis because of it's limitation of finding positive and negative class accurately, we have used other standard metrics on the datasets to validate the results.

Datasets	Classifier	Accuracy (%)	F1 Score		Precision		Recall		AUC
			Benign (%)	Malignant (%)	Benign	Malignant	Benign	Malignant	
Breast Cancer Wisconsin	Decision Tree	93.33	94.77	90.57	0.96	0.88	0.94	0.92	0.93
	Boosted Decision Tree	96.38	97.16	94.93	0.97	0.94	0.97	0.96	0.96

Figure 5.2: Performance analysis of Decision Tree and Boosted Decision Tree on Breast Cancer Wisconsin Dataset.

Figure 5.2 shows the experimental results that are found by applying Decision Tree and Boosted Decision Tree algorithms on Breast Cancer Wisconsin(original) dataset.

Here we can see that in every metrics, our proposed algorithm **Boosted Decision Tree** predicts cancer more effectively than **Decision Tree**. According to the results from figure 5.2, we can go to the conclusion that our proposed algorithm **Boosted Decision Tree** diagnose Breast cancer more accurately having fewest number of errors. This algorithm is very effective to predict the Breast cancer type and categories the patients whether the patients are in Benign Class or they are in Malignant Class.

Datasets	Classifier	Accuracy (%)	F1 Score		Precision		Recall		AUC
			Healthy Controls (%)	Patients (%)	Healthy Controls	patients	Healthy Controls	Patients	
Breast Cancer Coimbra	Decision Tree	67.06	63.97	68.15	0.65	0.69	0.64	0.69	0.66
	Boosted Decision Tree	72.94	70.57	75.05	0.72	0.74	0.71	0.75	0.73

Figure 5.3: Performance analysis of Decision Tree and Boosted Decision Tree on Breast Cancer Coimbra Dataset.

Figure 5.3 describes the experimental results that are found by applying Decision Tree and Boosted Decision Tree algorithms on Breast Cancer Coimbra dataset.

There are two classes in this Breast Cancer Coimbra dataset. One is Healthy Controls and the other is patients. From the figure 5.3 we can tell that our proposed algorithm **Boosted Decision Tree** predicts the two classes of Breast Cancer Coimbra dataset more accurately and effectively than **Decision Tree** because of their structures.

Datasets	Classifier	Accuracy (%)	F1 Score			Precision			Recall			AUC
			Class1 (%)	Class2 (%)	Class3 (%)	Class1	Class2	Class3	Class1	Class2	Class3	
Lung Cancer	Decision Tree	50.00	48.09	19.43	76.67	0.41	0.20	0.93	0.73	0.20	0.70	0.54
	Boosted Decision Tree	63.33	52.76	50.67	69.33	0.42	0.63	0.80	0.74	0.52	0.63	0.75

Figure 5.4: Performance analysis of Decision Tree and Boosted Decision Tree on Lung Cancer Dataset.

Figure 5.4 describes the experimental results that are found by applying Decision Tree and Boosted Decision Tree algorithms on Lung Cancer dataset.

There are three classes in this Lung Cancer dataset. They are the class1, class2, class3. From the figure 5.4 we see that our proposed algorithm **Boosted Decision Tree** gives much better accuracy as well as the AUC value than **Decision Tree**.

Again from figure 5.4 we can see that in terms of F1 Score, Precision and Recall, our proposed algorithm **Boosted Decision Tree** predicts Class1 and Class2 types of Lung Cancer more accurately and gives better performance than **Decision Tree**. But in terms of predicting Class3 type of Lung Cancer our proposed algorithm gives less accurate value than the Decision Tree.

Datasets	Classifier	Accuracy (%)	F1 Score		Precision		Recall		AUC
			No Recurrence (%)	Recurrence (%)	No Recurrence	Recurrence	No Recurrence	Recurrence	
Breast Cancer (Categorical)	Decision Tree	66.31	76.95	34.07	0.71	0.48	0.84	0.28	0.56
	Boosted Decision Tree	74.37	83.05	46.48	0.76	0.67	0.92	0.37	0.64

Figure 5.5: Performance analysis of Decision Tree and Boosted Decision Tree on Breast Cancer (Categorical) Dataset.

Figure 5.5 describes the experimental results that are found by applying Decision Tree and Boosted Decision Tree algorithms on Breast Cancer (Categorical) dataset.

There are two classes in this Breast Cancer (Categorical) dataset. One is No-recurrence and the other is Recurrence. From the figure 5.5 we can conclude that our proposed algorithm **Boosted Decision Tree** predicts the two classes of Breast Cancer (Categorical) dataset more accurately and effectively than **Decision Tree** because of their structures. Moreover, our proposed algorithm Boosted Decision Tree outperforms than the base line classifier Decision Tree for predicting the cancer types.

Boosting Algorithm is an Ensemble Learning Algorithm. In Ensemble Learning Algorithm multiple models come together to bring forth a model that is more effective. In every model, this algorithm overcomes the error of the previous model and finally gives a result which is more accurate. So, primarily it makes a sense that if we apply this algorithm on the performance of a weak learner it definitely improves the performance that we have for applying the weak learner on the dataset.

Based on this properties of the Ensemble Learning Algorithm, we have planned to use this on the Cancer datasets to predict the Cancer types in early stages. If we predict cancer in early stages, it will reduce the mortality rate of cancer death. This is the main issue of our research work.

We have also planned to use this Boosting Algorithm as our proposed algorithm named **Boosted Decision Tree** because of it's structures of predicting many diseases effectively. Moreover, we have taken boosting algorithm as our proposed algorithm because it is not applied previously on this cancer related data field and also belonging the properties that can help to diagnose cancer diseases having the fewest number of errors.

So, finally we have observed from the figure 5.2, 5.3, 5.4, 5.5 that for all of the metrics, our proposed algorithm **Boosted Decision Tree** predicts different cancer types more effectively than **Decision Tree** when applied on all of these four cancer datasets. So, the results are matched as we expect in our primary considerations.

Conclusion

In this chapter, we have showed the result, comparison and experimental analysis of our proposed algorithm on different cancer related datasets. It can conclude that our proposed algorithm named **Boosted Decision Tree** is very effective to diagnose cancer having the fewest number of errors with compared to **Decision Tree** approach because of it's properties and able to reduce the mortality rate of death for cancer.

Chapter 6

FUTURE WORKS AND CONCLUSION

Introduction

Future works and Conclusion are important issues of a research work. In this section we have described the overall conclusions that we have achieved through our thesis and what would be done in near future.

6.1 Future Works

As our proposed approach is not applied before on this cancer related data field and has been investigated on the numerical and categorical datasets, this approach can be used on other types of cancer datasets to determine the effect in near future.

Conclusion

In this paper, Decision Tree is treated as a ‘weak learner’ to generate it as a strong learner by applying Adaboost algorithm. In every model this algorithm overcomes the error of the previous model and finally gives a result which is more accurate. So, different measurements on the datasets remarkably specify that Boosted Decision Tree outperforms than Decision Tree as stated in the above. Moreover, Boosted Decision Tree has enhanced the performance with the effective accuracy of predicting cancer types for each of the four datasets. This model gives an effective performance for predicting both the binary class as well as the multi-class cancer types where Decision Tree finds a limitation. But this approach is restricted when the

no. of instances of the dataset is small while having a higher dimensionality. Preprocessing the datasets can be an interesting extension of this work with appropriate feature selection techniques before progressing to Boosted Decision Tree algorithm.

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